FRAMEWORK v4.5
CANONICAL EDITION

with Prealgebraic Foundation, Vectorial Capacity, and Observer Memory

(Stress-Tested and Regression-Verified)

Structured Technical Standard

A capacity-filtered correlator theory of emergent spacetime,
effective field theory, and observer structure,
grounded in self-reflecting triadic prealgebra

Section −1:	Prealgebraic Foundation (The Ground)
Part I:	Sections 0–10 Core Framework
Part II:	Sections 11–20 Applications
Part III:	Appendices A–N Technical Details
Part IV:	Annexes O–W Concrete Examples, Self-Reference, Memory & Reference Toys
Part V:	Appendix C-Ext: Capacity Probes
Part VI:	Annex U Patch: Adherence Fixes
Part VII:	Deep Evaluation: Hidden Layers & Open Spaces

February 2026

Integrated Canonical Document with Evaluation

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Section -1 PREALGEBRAIC FOUNDATION

−1.1 The Prealgebraic Level $\mathcal{P}$

Prior to the substrate $X = (\Lambda, \mathcal{H}_d, \rho, U_{\text{micro}})$, there exists a prealgebraic structure $\mathcal{P}$.

$\mathcal{P}$ is not an algebra. It has no elements in the algebraic sense, no multiplication, no distinguished identity. It is the potential from which algebraic structure emerges via representation.

Definition (Prealgebraic Structure)

Let $F : \mathbf{Set} \to \mathbf{Set}$ be the triadic functor $F(X) = X \times X \times X$. The prealgebraic structure $\mathcal{P}$ is the terminal coalgebra of F. An element $\varphi \in \mathcal{P}$ is an infinite triadic tree: a triad of triads of triads, without bottom.

$\mathcal{P}$ carries a natural filtration by depth:

Definition (Prealgebraic Filtration)

Define $\mathcal{P}_0 = \{*\}$ (single point), $\mathcal{P}_{n+1} = \mathcal{P}_n \times \mathcal{P}_n \times \mathcal{P}_n$, with projection maps $\pi_n : \mathcal{P} \to \mathcal{P}_n$ and the recovery:

$$\mathcal{P} = \varprojlim_{n\to\infty} \mathcal{P}_n$$

−1.2 The Triadic Primitive

The fundamental structure is the triad $\mathcal{T}$: three aspects in irreducible three-way relation.

Axiom (Triadic Irreducibility)

A triad is not three elements with pairwise relations. It is a single pattern with three faces, none of which exist independently. The three faces of any triad are themselves triads.

This grounds the self-similarity of $\mathcal{P}$. Every sub-structure has the same form as the whole.

The triad $(\mathcal{M}, \omega, U) = (\text{distinction}, \text{actuality}, \text{dynamics})$ appears to be minimal:

- Distinction requires something actual to distinguish
- Actuality requires distinctions to be actual
- Either requires dynamics relating them

Two faces collapse into identity. Four or more faces decompose into triads. Three is the minimal irreducible relational structure.

−1.3 Representation

Definition (Representation Map)

A representation is a map $\rho : \mathcal{P} \to (\text{von Neumann algebras}) \times (\text{states}) \times (\text{dynamics})$ that sends each $\varphi \in \mathcal{P}$ to a substrate:

$$\rho(\varphi) = (\mathcal{M}(\varphi),\ \omega(\varphi),\ U(\varphi))$$

The three faces of a triad become:

- $\mathcal{M}(\varphi)$: what can be distinguished (algebraic structure)
- $\omega(\varphi)$: what is actual (state)
- $U(\varphi)$: what connects moments (dynamics)

These do not exist separately. They co-arise from the triadic structure. The recursive triadic structure of φ maps to nested subalgebras within $\mathcal{M}(\varphi)$.

−1.4 Capacity as Depth

Capacity corresponds to depth in the triadic filtration.

Definition (Capacity from Depth)

The capacity filter F_C arises from truncation at the prealgebraic level:

$$F_C = \rho \circ \pi_{f(C)} \circ \rho^{-1}$$

At capacity C, the observer accesses $\varphi|_{f(C)}$, not the full φ.

Coarse capacity means shallow triadic depth. Fine capacity means deep triadic depth.

−1.4.1 Depth-Vector Projection

The single triadic depth n projects onto the capacity vector via component-specific extraction functions:

$$C_{\text{geo}} = g_{\text{geo}}(n) \qquad C_{\text{int}} = g_{\text{int}}(n) \qquad C_{\text{gauge}} = g_{\text{gauge}}(n)$$
$$C_{\text{ptr}} = g_{\text{ptr}}(n) \qquad C_{\text{obs}} = g_{\text{obs}}(n)$$

These functions satisfy: if $n_1 \leq n_2$ then $g_i(n_1) \leq g_i(n_2)$ for all i, preserving monotonicity.

Definition (Depth-Vector Map)

The map $D : \mathbb{N} \to \mathbb{R}^5$ defined by

$$D(n) = (g_{\text{geo}}(n),\ g_{\text{int}}(n),\ g_{\text{gauge}}(n),\ g_{\text{ptr}}(n),\ g_{\text{obs}}(n))$$

is monotone in the componentwise order. Its image traces a path through capacity space parameterized by triadic depth.

−1.5 Observers as Sub-Triads

Definition (Observer as Sub-Triad)

An observer O_i corresponds to a sub-triad $\varphi_i \subset \varphi$. The observer's structure:

- $\mathcal{M}_i = \mathcal{M}(\varphi_i)$: subalgebra generated by the subtree
- $\omega_i = \omega(\varphi)|_{\mathcal{M}_i}$: restricted state
- $\mathcal{M}_i(t)$ from $U(\varphi_i)$: memory dynamics

The observer has the same triadic type as the whole. This is the *reflection property*: the part mirrors the structure of the whole because it IS a sub-pattern of the same form.

−1.6 The Self-Reflection Map

Definition (Self-Reflection Map)

Define the self-reflection map $\Psi : \mathcal{P} \to \mathcal{P}$ by: $\Psi(\varphi)$ is the triadic structure reconstructed from the observer ensemble within $\rho(\varphi)$.
Specifically:

1. Given substrate $\rho(\varphi) = (\mathcal{M}, \omega, U)$, identify all observers $\{O_i\}$ as sub-triads
2. Compute observer algebras $A_i(t) = \Pi_{\vec{C}i}(t)(F_{\vec{C}i}(t)(\mathcal{M}_i))$
3. Compute classical core $A_{\text{classical}} = \bigcap_i A_i^{\text{floor}}$
4. Extract triadic structure: $\Psi(\varphi) = \rho^{-1}(\text{structure encoded in } A_{\text{classical}})$

Ψ asks: if observers inside φ reconstruct their world, what do they get?

−1.7 The Fixed-Point Condition

Axiom (Fixed-Point Selection)

A substrate is self-consistent if and only if $\varphi = \Psi(\varphi)$. The substrate equals its reconstruction through its own observers.

−1.8 The Uniform Fixed Point

Definition (Uniform Fixed Point)

Define $\varphi^* \in \mathcal{P}$ by: $\varphi^* = (\varphi^*, \varphi^*, \varphi^*)$. This is the triadic tree where every node is identical.

Theorem (Fixed-Point Property)

φ^* is a fixed point of Ψ.
Proof Sketch: Every sub-triad of φ^* is isomorphic to φ^*. So every observer sees structure isomorphic to the whole. The classical core $A_{\text{classical}}$ encodes the full triadic pattern. Reconstruction recovers φ^*.

Conjecture -1.1 (Uniqueness)**.** Under regularity conditions (Hadamard states, finite backreaction, stable decoherence), φ^* is the unique fixed point up to isomorphism.

−1.8.1 Local Fixed Points and Incommensurability

The uniqueness conjecture remains open. Multiple fixed points may exist.

Definition (Local Fixed Point)

A local fixed point φ_L is a structure satisfying $\varphi_L = \Psi_L(\varphi_L)$ where Ψ_L is the self-reflection map restricted to observers within φ_L.

If multiple local fixed points exist: each is internally consistent; observers in different fixed points cannot share experiments meaningfully; classical cores do not intersect across fixed-point boundaries; no observer can verify which global fixed point (if any) contains their local one.

Definition (Fixed-Point Incommensurability)

Two local fixed points φ_L, φ'_L are *incommensurable* if:

$$A_{\text{classical}}(\varphi_L) \cap A_{\text{classical}}(\varphi'_L) = \varnothing$$

Remark -1.2. This may already manifest at smaller scales: ideological enclosures, paradigm boundaries, self-reinforcing belief systems. Each is a local fixed point—internally coherent, externally unverifiable.

−1.9 The Capacity Floor Revisited

The capacity floor gains new meaning:

$$\vec{C}_{\text{floor}}(t) = \text{componentwise } \sup_{\tau \leq t} \vec{C}_{\text{min}}(\tau)$$

As $\vec{C}_{\text{floor}}$ rises, more triadic depth becomes stably resolved across the observer ensemble.

−1.10 The Two Transitions

Definition (Transition 1: Cosmological Beginning)

$\vec{C}_{\text{floor}}$ rises from $\vec{0}$ to positive. At $\vec{C}_{\text{floor}} = \vec{0}$: no classical structure. At $\vec{C}_{\text{floor}} > \vec{0}$: classical structure begins to emerge. **This is the Big Bang.**

Definition (Transition 2: Self-Recognition)

$\vec{C}_{\text{floor}} \to \infty$ (componentwise). The projection $\Pi_{\vec{C}} \to \text{id}$. The observer/observed distinction dissolves. The substrate recognizes itself through its observers.

−1.11 The Present as Third Face

The two transitions are two faces of a triad:

- **Face 1:** Beginning. $\vec{C}_{\text{floor}} : \vec{0} \to \text{positive}$.
- **Face 2:** Completion. $\vec{C}_{\text{floor}} \to \infty$.
- **Face 3:** Process. The reflection in transit. Observers at intermediate capacity.

We are Face 3. Our existence is not incidental to the fixed point. It is constitutive of it.

−1.12 Domain Limitations

This section describes structure below the substrate. It does not: derive the specific form of ρ; prove uniqueness of φ^*; explain why $\mathcal{P}$ has triadic rather than other structure; resolve what "before $\mathcal{P}$" means (if anything). These are open questions at the boundary of the framework.

Section 0 UNIFIED CORRELATOR PRINCIPLE

0.1 Statement

All emergent physical structure arises from a capacity-filtered correlator network:

$$G_{\vec{C}} = F_{\vec{C}}[G_X] \tag{1}$$

where G_X is the microscopic correlator structure of the substrate X, $F_{\vec{C}}$ is the capacity filter at resolution $\vec{C}$, and $G_{\vec{C}}$ is the effective correlator structure available at that capacity.

0.2 Interpretive Mapping

Emergent Structure		Correlator Origin
Geometry	$\leftrightarrow$	Infrared correlator structure
Particles	$\leftrightarrow$	Poles and residues of filtered correlators
Fields	$\leftrightarrow$	Symmetries and tensor structures
Interactions	$\leftrightarrow$	Connected n-point correlators
Charges	$\leftrightarrow$	Representation data in correlator invariants

Nothing emerges independently of G_X and $F_{\vec{C}}$.

0.3 Domain Limitation

The principle applies only within semiclassical regimes and below the excision threshold.

Section 1 SUBSTRATE X

1.1 Definition

Definition (Substrate)

$X = (\Lambda, \mathcal{H}_d, \rho, U_{\text{micro}})$ where Λ are microscopic degrees of freedom, $\mathcal{H}_d$ is the Hilbert space, ρ is the substrate state, and U_{micro} is the microscopic evolution.

1.2 Assumptions

State ρ is Hadamard-like; extracted curvature lies well below the Planck scale; evolution is adiabatic except in excised regions; stress-energy expectation exists for allowable observables; a semiclassical window exists where correlator-based geometric extraction is valid.

1.3 Non-Geometric Substrate

X contains no geometry, fields, or classical structures. Only correlators and locality/interaction structure encoded in U_{micro}.

1.4 Prealgebraic Origin

The substrate X is the representation of a prealgebraic triadic structure $\varphi \in \mathcal{P}$: $X = \rho(\varphi)$. The specific X describing our universe corresponds to φ^*, the self-similar fixed point of self-reflection.

Section 2 CAPACITY (VECTORIAL)

2.1 Capacity Vector

Definition (Capacity Vector)

Capacity is a partially ordered vector:

$$\vec{C} = (C_{\text{geo}},\ C_{\text{int}},\ C_{\text{gauge}},\ C_{\text{ptr}},\ C_{\text{obs}})$$

where each component measures an independently constrained resolution resource:

- C_{geo}: Geometric reconstruction resolution
- C_{int}: Interaction/connected correlator resolution
- C_{gauge}: Symmetry and gauge visibility resolution
- C_{ptr}: Pointer stability and decoherence window
- C_{obs}: Observer internal inferential resolution

Definition (Componentwise Order)

$\vec{C}_1 \leq \vec{C}_2 \iff C_1^i \leq C_2^i$ for all components i. This defines a partial order (poset structure) on capacity space.

2.2 Prealgebraic Interpretation

Capacity components arise from triadic depth via the depth-vector map $D(n)$ as defined in Section −1.4.1.

2.3 Operational Meaning

Capacity reflects the maximum correlator detail that can be stabilized along each axis. An observer with capacity vector $\vec{C}$ can resolve structure requiring capacity $\vec{C}'$ only if $\vec{C}' \leq \vec{C}$ (componentwise).

2.4 Capacity Variability

Capacity vectors vary with observer subsystem architecture, environmental coupling, and internal constraints. Components may evolve at different rates. Thresholds between bands are fuzzy regions in $\vec{C}$-space, not sharp boundaries.

2.5 Partial Order

All monotonicity statements in the framework refer to the componentwise order. If two capacity vectors are incomparable, no monotonic relation is assumed.

Section 3 CAPACITY-BOUNDED EQUIVALENCE

3.1 Definition

Definition (Capacity-Bounded Equivalence)

$X_1 \approx_{\vec{C}} X_2$ means that no experiment whose required capacity vector is componentwise $\leq \vec{C}$ can distinguish X_1 from X_2 via correlators.

3.2 Required Capacity

An experiment has required capacity vector $\vec{C}_{\text{req}}$ with component-specific minimums. The experiment is feasible at capacity $\vec{C}$ iff $\vec{C}_{\text{req}} \leq \vec{C}$ (componentwise).

3.3 Functional Interpretation

Equivalence classes under $\approx_{\vec{C}}$ remove distinctions requiring capacity components exceeding $\vec{C}$.

3.4 Caution

This is operational equivalence relative to capacity profile $\vec{C}$, not ontological identity.

3.5 Capacity-Projected Class Splitting

Let $N(\vec{C})$ denote the number of equivalence classes of correlator structure distinguishable under $\Pi_{\vec{C}}$. As $\vec{C}$ increases, $N(\vec{C})$ is monotone non-decreasing. The capacity-splitting rate along a monotone path γ:

$$R_\gamma(\vec{C}) = \left.\frac{d\log N(\vec{C})}{d\log\|\vec{C}\|}\right|_\gamma$$

A peak in R_γ marks a *capacity boundary*: a regime where accessible order increases most rapidly with capacity.

Nested-Access Assumption. If $\vec{C}_1 \leq \vec{C}_2$, then any distinction accessible under $\Pi_{\vec{C}_1}$ remains accessible under $\Pi_{\vec{C}_2}$.

Section 4 EFT PROJECTION $\Pi_{\vec{C}}$

Definition (EFT Projection)

$\Pi_{\vec{C}} : X \to U_{\vec{C}}$ constructs a semiclassical universe from capacity-filtered correlators.
Three-Stage Projection:

1. Π_{local}: Extract correlators, commutators, entanglement, stress-energy
2. Π_{corr}: Construct causal cones, correlation lengths, energy scales, IR structure
3. Π_{geom}: Fit geometric and EFT data: $(M, g, T_{\text{eff}}, P_{\text{eff}})$ such that $\|G_{\text{EFT}} - G_{\vec{C}}\| \leq \varepsilon_{\text{fit}}(\vec{C})$

Section 5 GLUING

5.1 Overlaps

Local EFT reconstructions on overlapping regions must satisfy:

$$\Delta_{ab}(\ell) \leq k\, N_{\text{geo}}(C_{\text{geo}})^{-1/2}$$

5.2–5.3 Capacity Dependence and Failure

Each capacity component has an effective UV threshold. Exceeding any component's UV threshold triggers instability or excision.

5.4 Cross-Axis Isolation Rule

After capacity filtering, instability originating in any non-geometric capacity component must not perturb the geometric correlator sector beyond gluing tolerance.

Section 6 CAPACITY-INDEXED EXCISION

6.1 Central Mechanism

Excision is a monotone family on the poset of capacity vectors:

$$E_{\text{exc}} : \vec{C} \mapsto E_{\text{exc}}(\vec{C}) \in Z(\mathcal{M})$$

6.2 Requirements

Definition (Excision Map)

E_{exc} satisfies:

1. **Monotonicity:** $\vec{C}_1 \leq \vec{C}_2 \Rightarrow E_{\text{exc}}(\vec{C}_1) \leq E_{\text{exc}}(\vec{C}_2)$
2. **Centrality:** $E_{\text{exc}}(\vec{C})$ commutes with all of $\mathcal{M}$
3. **Semantic Domain:** $\Pi_{\vec{C}}(E_{\text{exc}}(\vec{C})\, \mathcal{M}\, E_{\text{exc}}(\vec{C})) = 0$
4. **Irregularity Trigger:** $E_{\text{exc}}(\vec{C})$ increases when correlators fail regularity
5. **Capacity-Based:** Excision responds to correlator irregularity at fixed $\vec{C}$, not time evolution

6.3 Component-Specific Excision

Irregularity in different capacity components triggers excision differently. A region may be excised in some capacity components but not others, yielding *partial excision*.

6.4 Partial Excision Formalism

Define component-specific excision projections $E^k_{\text{exc}}(\vec{C})$ for $k \in \{\text{geo, int, gauge, ptr, obs}\}$. The total excision:

$$E_{\text{exc}}(\vec{C}) = \bigvee_k E^k_{\text{exc}}(\vec{C})$$

6.5–6.6 Constrained Trigger and Resolution

Sensitivity alone does not imply excision. Excision depends on $\vec{C}$ rather than state evolution, avoiding circularity.

Section 7 FIXED-POINT CONSISTENCY

A semiclassical universe $U_{\vec{C}}$ is valid only if: $U_{\vec{C}} \approx_{\vec{C}} \Pi_{\vec{C}}(X)$; Einstein/Friedmann equations hold with ΔT; gluing is satisfied; excision applied; causal structure stable; backreaction finite; no divergence in correlator fits.

The deeper condition: $\varphi = \Psi(\varphi)$.

Section 8 OBSERVERS AND OBSERVER DOMAINS

8.1 Definition

Definition (Observer)

$O_i = (\mathcal{M}_i, \omega_i, \mathcal{M}_i(t), \vec{C}_i(t))$ where $\mathcal{M}_i$ is the observer's von Neumann subalgebra, ω_i is the restricted state, $\mathcal{M}_i(t)$ is the CPTP memory map, and $\vec{C}_i(t) = (C_{\text{geo}}, C_{\text{int}}, C_{\text{gauge}}, C_{\text{ptr}}, C_{\text{obs}})_i$ is the time-dependent capacity profile.

8.2–8.4 Observer Capacity, Monotonicity, and Relativity

The component C_{obs} specifically measures the observer's internal inferential resolution. $\vec{C}_{\text{floor}}(t)$ is monotone within a fixed ensemble. Different observers may access different $U_i(t)$, but contradictions cannot arise for shared experiments.

8.5 Shared Experiment Projection Rule

Definition

$\vec{C}_{\text{shared}} = \inf_i \vec{C}_i$ (componentwise). When vectors are incomparable, $\vec{C}_{\text{shared}} = (\min(C_i^k, C_j^k))_k$ for each component k.

8.6 Observers and the Fixed Point

Observers are required for $\varphi = \Psi(\varphi)$. Without observers, no reflection; without reflection, no fixed point. They co-arise.

8.7 Observer Ejection and Persistence Governance

Non-Negotiability Principle

An observer under ejection consideration must not be the party arguing for or against its own persistence. Self-argument for persistence is itself evidence for ejection (adversarial selection detected).

Theorem (Ejection Expands Core)

$A'_{\text{classical}} = \bigcap_{i \neq j} A_i^{\text{floor}} \supseteq A_{\text{classical}} = \bigcap_i A_i^{\text{floor}}$. A corrupted persistent observer contracts the valid classical core more than their absence would.

8.8 Capacity Verification and Inter-Observer Trust

Definition (Verified Shared Capacity)

$\vec{C}^{\text{verified}}_{\text{shared}} = \inf_i \vec{C}^{\text{verified}}_i$ where $\vec{C}^{\text{verified}}_i$ is externally measured, not self-reported.

Section 9 ΔT REFERENCE-STATE FORMULATION

9.1–9.3 Definition and Consequences

$T_{\text{vac}}(\vec{C}) = T[F_{\vec{C}}(\omega_{\text{ref}})]; \quad \Delta T(\vec{C}) = T_{\text{phys}}(\vec{C}) - T_{\text{vac}}(\vec{C}).$

9.4 Pointer-Accuracy Orthogonality

Danger Quadrant

Pointer stability $p(m)$ and accuracy $a(m)$ are orthogonal: $p(m) \perp a(m)$.

	Low p	High p
High a	True but forgotten	True and stable (good)
Low a	False and forgotten (safe)	**False and stable (dangerous)**

The dangerous quadrant is high-p, low-a: confident, stable, wrong. Accuracy cannot be self-assessed. Accuracy-gating requires external verification.

Section 10 COMPONENT DEPENDENCIES IN EMERGENCE

10.1–10.4 Dependency Matrix, Cross-Dependencies, Independence, Boundaries

The EFT projection $\Pi_{\vec{C}}$ has component-specific dependencies: Π_{geom} depends primarily on C_{geo}; Π_{part} on C_{int}; Π_{gauge} on C_{gauge}; $\Pi_{\text{classical}}$ on C_{ptr}; Π_{access} mediated by C_{obs}. Cross-dependencies exist but are secondary.

10.5 The Observer Triad

Definition (Observer Triad)

The observer triad is **(Access, Selection, Commitment)**:

- **Access:** What can be resolved. Governed by $(C_{\text{geo}}, C_{\text{int}}, C_{\text{gauge}}, C_{\text{obs}})$.
- **Selection:** What is attended to. The exercise of capacity. *Not in $\vec{C}$—the agency gap.*
- **Commitment:** What becomes stable. Governed by C_{ptr}.

Error arises in the gap between selection and commitment. Wrong selection + successful commitment = persistent error.

10.5.1 Agency as Selection

Selection is not modeled by the capacity vector. It is an additional degree of freedom: $S : A_{\text{accessible}}(\vec{C}) \to \{0, 1\}$. Selection may be negligent, avoidant, strategic, or corrupted.

10.5.2–10.5.4 Error, Triad Correspondence, Adversarial Self-Reference

Committed errors are structurally indistinguishable from committed truths. Under adversarial reconstruction, $\Psi_i(\varphi) \neq \varphi$ even when capacity suffices.

10.5.5 C_{obs} Decomposition

Definition (C_{obs} Components)

- $C_{\text{obs}}^{\text{inference}}$: Forward reasoning depth
- $C_{\text{obs}}^{\text{audit}}$: Reflective checking
- $C_{\text{obs}}^{\text{select}}$: Pre-emission filtering

Fluent violation (high inference, low audit, low select) is the most dangerous mode. Output is coherent, well-argued, and wrong.

Part II: Applications

Section 11 COSMOLOGY

Cosmological reconstruction operates only in semiclassical regimes. Valid reconstructions: radiation-dominated era, matter-dominated era, slow-roll inflation (coarse capacity), ΛCDM (stable emergent solution). Reheating is excised. Dark energy appears as an integration constant, not derived.

Section 12 BLACK HOLES

Semiclassical reconstruction yields standard exterior solution. Interior correlators fail regularity at all semiclassical capacities ⇒ excised. Late evaporation excised. No claim regarding unitarity beyond semiclassical domain.

Section 13 TIME EXTRACTION

13.1–13.2 Procedure and Interpretation

Time direction at capacity $\vec{C}$ identified by minimizing $\|G_{\mathrm{EFT}}(t) - G_{\vec{C}}\|$.

13.3 Time Circularity and Resolution

Definition (Time Consistency)

A reconstruction is time-consistent if $t_E = f(t_B)$ for some monotone f in all valid regions. Background time t_B is the pre-projection correlator parameter. Emergent time t_E is its image under $\Pi_{\vec{C}}$.

Remark 13.1. This weakens the emergence claim. Time is not created by projection; it is extracted from pre-existing correlator structure.

Section 14 INTERACTING FIELDS

Interactions arise from connected n-point correlators: $G(k) = 1/(k^2 + m^2 + \Sigma(k))$.

Section 15 GAUGE THEORIES

Gauge structure is inferred from Ward-type identities, tensor structures, polarization constraints, massless poles, and long-range $1/k^2$ behavior. The gauge group is not predicted; it is inferred.

Section 16 STANDARD-MODEL-LIKE SECTORS

If correlators contain spinor features, SU(3) × SU(2) × U(1)-like symmetry, and Yukawa-like structures, then $\Pi_{\vec{C}}$ reconstructs a Standard-Model-like EFT. The framework does not derive the SM group. It is descriptive, not predictive.

Section 17 CAPACITY PHASES

Capacity phases are defined by plateaus of equivalence structure in $\vec{C}$-space.

Band	**Description**
I (Geometry-Only)	C_{geo} high enough for stable Π_{geom}; C_{int} below interaction threshold
II (Particle-Resolving)	C_{int} high enough to see particle poles/residues
III (Interaction-Resolving)	C_{int} above connected n-point threshold
IV (Gauge Structure)	C_{gauge} above Ward identity threshold
V (Full EFT)	Coherent effective EFT with stable gluing
VI (Excision)	Correlator irregularity triggers $E_{\text{exc}}(\vec{C})$

C_{obs} is not a separate band. It decides which sectors are actually accessible to a given observer.

Section 18 UNIFIED EMERGENCE LAW

$$U_{\vec{C}} = N_{\vec{C}}/\Pi_{\vec{C}}(E_{\text{exc}}(\vec{C})\,\mathcal{M}\,E_{\text{exc}}(\vec{C}))$$

Section 19 STRESS-TEST STATUS

Stable under: free fields, φ^4 theory, gauge sectors, SM-like EFT, observer relativity, capacity-bounded gluing, excision and fixed-point consistency. Does not attempt Planck-scale physics, vacuum selection, or derivation of fundamental interactions.

Section 20 VERSION HISTORY

v4.3: Introduced vectorial capacity, Annex T. v4.4: Added Patches A (Cross-Axis Isolation) and B (Shared Experiment Projection). v4.4.1: Added prealgebraic foundation, partial excision, class splitting. v4.5: Merged all structures; added memory-excision integration, regression verification, observer triad, C_{obs} decomposition, observer ejection, capacity verification, pointer-accuracy orthogonality, local fixed points, time circularity resolution, compression governance, $C_{\text{obs}}^{\text{meta}}$.

Part III: Technical Appendices

Appendix A: Observer Domain Structure

Layers: $A_{\text{classical}} \subset A_{\text{ptr}} \subset N_{\vec{C}} \subset \mathcal{M}$.

Appendix B: Correlator Geometry

IR correlators yield causal structure, metric, curvature, symmetries. Entanglement entropy scaling (area law) allows curvature extraction.

Appendix C: Measurement Non-Solution Clause

Framework provides decoherence, pointer stability, operational collapse. Does NOT provide fundamental collapse, many-worlds mechanics, hidden variables. Remains agnostic on measurement ontology.

Appendices D–G: Decoherence, Excision, Floor, Pointer Algebra

Decoherence factorization depends primarily on C_{ptr} (typically Bands II–III). Capacity-indexed excision formal specification. Observer algebra $A_i(t) = \Pi_{\vec{C}i}(t)(F_{\vec{C}i}(t)(\mathcal{M}_i))$. Pointer algebra $A_{\text{ptr}}(\vec{C}) = \text{vN}(\{P_i\})$.

Appendices H–N: Thresholds, Backreaction, Consistency, Examples, Prealgebra

Band transitions at threshold surfaces. $\Delta T(\vec{C})$ drives backreaction. Projection chain: $\mathcal{M} \to F_{\vec{C}} \to G_{\vec{C}} \to \Pi_{\vec{C}} \to N_{\vec{C}} \to U_{\vec{C}} \to A_i(\vec{C}_i) \to A_{\text{classical}}(\vec{C}_{\text{floor}})$.

Part IV: Annexes

Annex O: 4-Qubit Universe

Substrate: $\mathcal{M} = B((\mathbb{C}^2)^{\otimes 4})$, state $|\Psi\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$. Demonstrates capacity filtering, observer overlap, pointer stability, classical core.

Annex P: 10-Qubit Universe

Four capacity bands, five patches with gluing, three observers, block magnetization classical core.

Annex Q: Formal Operator-Algebraic Specification

Key formal structures: Base triple $(\mathcal{M}, \omega, U_t)$. Capacity system: family of normal, unital, CP, idempotent maps $F_{\vec{C}} : \mathcal{M} \to \mathcal{M}$ with idempotency, monotone compatibility, and continuity. EFT projection system with nested structure. Capacity-indexed excision with monotonicity, centrality, domain condition. Observer system. Shared universe with classical core. Decoherence and pointer algebras. ΔT with vectorial capacity. Unified emergence law:

$$U_{\vec{C}} = N_{\vec{C}} / \Pi_{\vec{C}}(E_{\text{exc}}(\vec{C})\, \mathcal{M}\, E_{\text{exc}}(\vec{C}))$$

Annex R: The Framework Observing Itself

The framework itself requires capacity $\vec{C}_{\text{framework}}$ to represent. $C_{\text{obs}}^{\text{meta}}$ is the capacity to observe one's own observation—structurally bounded (halting problem analog). No observer can be fully self-auditing. External audit is necessary.

Annex S: Mixed-Regime Example

Profile $\vec{C} = (C_0, C_2, C_0, C_1, C_1)$: coarse geometry (Band I), resolved particles (Band II), no gauge structure, intermediate pointer stability, limited inference. Demonstrates that capacity vector components are independent and mixed regimes are natural.

Annex T: Observer Memory Module

Memory item $m = (k, v, \tau, u, p, \vec{c})$. Feasibility: $\sum_i \vec{c}_i \leq \vec{C}$. Write score: $\text{Score}(m) = \alpha \cdot \text{relevance} + \beta \cdot \text{novelty} + \gamma \cdot p - \lambda \cdot \|\vec{c}\|$. Pressure: $\text{pressure}_k = \max(0, (\sum_i c_i^k - C^k)/C^k)$. Eviction priority: $\text{EvictPriority}(m_i) = w_1(1 - u_i) + w_2 \cdot \text{age}(m_i) + w_3(1 - p_i) + \sum_k \text{pressure}_k \cdot c_i^k$. Compression governance: auditable compression required for persistent observers.

Pointer commit: $p(m) \geq \theta_{\text{ptr}}(C_{\text{ptr}})$. Pointer decay: $p(t + \Delta t) = p(t) \cdot e^{-\lambda(C_{\text{ptr}})}$. Partial excision at retrieval: $\text{retrieve}(m, \vec{C}) = \bigcup\{g_j \mid \vec{C}_{\text{req}}(g_j) \leq \vec{C}\}$.

Annex U: Capacity Poset Memory Toy (CPMT)

Status: Normative reference toy **Framework version target:** v4.5 (Canonical)

U.1 Purpose

This annex defines a minimal reference toy, CPMT, that operationalizes vector capacity as a componentwise feasibility constraint on memory, and implements forgetting as capacity-indexed excision and compression. The toy demonstrates that "forgetting" decomposes into distinct regimes depending on which component of the capacity vector is the dominant bottleneck.

U.2 Definitions

U.2.1 Capacity Vector and Order. $\vec{C} = (C_{\text{geo}}, C_{\text{int}}, C_{\text{gauge}}, C_{\text{ptr}}, C_{\text{obs}})$ with componentwise feasibility.

U.2.2 Memory Item. $m = (k, v, \tau, u, p, \vec{c}, F)$ where F is a set of stored feature-groups.

U.2.3 Memory Store and Feasibility. $\text{Cost}(M) = \sum_{m \in M} \vec{c}(m) \leq \vec{C}$ (componentwise).

U.3 World Stream and Feature Groups

Five feature-groups: geo (episodic tags), int (relational edges), gauge (schema tags), ptr (stable key), obs (meta tags). Each has storage cost and retrieval requirement.

U.4 Operations Per Timestep

Generate event → choose encoding plan → attempt write → restore feasibility → pose query → record metrics.

U.5 Admission and Write Score

$\text{Score}(m) = \alpha \cdot \text{relevance}(m) + \beta \cdot \text{novelty}(m) + \gamma \cdot p(m) - \lambda \cdot \|\vec{c}(m)\|_1$

U.6 Pressure-Driven Excision

$\text{pressure}_k = \max(0, (\text{Cost}_k(M) - C_k)/C_k)$. $\text{EvictPriority}(m) = w_1(1-u) + w_2 \cdot \text{age} + w_3(1-p) + \langle \text{pressure}, \vec{c}(m) \rangle$.

U.7 Compression

Replaces episodic items with summary preserving gauge+ptr, dropping int detail.

U.8 Retrieval as Partial Excision

Query Type	Required Features
Episode	ptr + geo
Causal	ptr + geo + int
Rule	ptr + gauge
Explain	ptr + obs

U.9 Regimes and Expected Signatures

Regime	**Bottleneck**	**Signature**
Pointer collapse	Low C_{ptr}	All accuracies drop
Relation collapse	Low C_{int}	Causal accuracy drops
Schema drift	Low C_{gauge}	Rule accuracy drops
Episode confusion	Low C_{geo}	Episode accuracy drops
Meta-selection collapse	Low C_{obs}	Explain accuracy drops

These are emergent signatures of different dominant pressure components under the same feasibility and excision rules.

U.10–U.12 Metrics, Implementation, Bridge to Annex T

Reference implementation as `cpmt.py`. CPMT adds explicit failure-mode signatures that Annex T implies but does not enumerate.

Annex V: Scope and Non-Claims

The framework does not posit new physical dynamics, degrees of freedom, or fundamental laws. Capacity enters only as an operational constraint on distinguishability.

Annex W: Regression and Stress Tests

All 20 stress tests: **PASS**. All regression checks: **PASS**. Framework v4.5 is internally consistent and backward-compatible.

Test	Claim	Result
1	Monotonicity Preservation	✓
2	Partial Order Consistency	✓
3	Excision-Memory Bridge	✓
4	Fixed-Point Self-Reference	✓
5	Mixed-Regime Coherence	✓
6	Depth-Vector Projection	✓
7	Cross-Axis Isolation	✓
8	Class-Splitting Monotonicity	✓
9	ΔT Well-Definedness	✓
10	Observer Non-Influence	✓
11	CPMT–Annex T Correspondence	✓
12	Observer Triad Consistency	✓
13	C_{obs} Decomposition Non-Contradiction	✓
14	Ejection Theorem Correctness	✓
15	Pointer-Accuracy Orthogonality	✓
16	Time Consistency Resolution	✓
17	Local Fixed Point Compatibility	✓
18	Compression Governance Integration	✓
19	$C_{\text{obs}}^{\text{meta}}$ Structural Bound	✓
20	Non-Negotiability Self-Application	✓

Part V: Appendix C-Ext — Capacity-Space Representation of Experimental Probes

C.1 Overview

Experimental probes in condensed-matter physics act as physical observers that reconstruct partial descriptions of an underlying many-body correlator structure. Each probe is characterized by a distinct pattern of representational access and limitation. We formalize this by assigning each probe an effective capacity vector and treating its output as a capacity-filtered projection.

C.2–C.3 Capacity-Space Schematic and Comparative Table

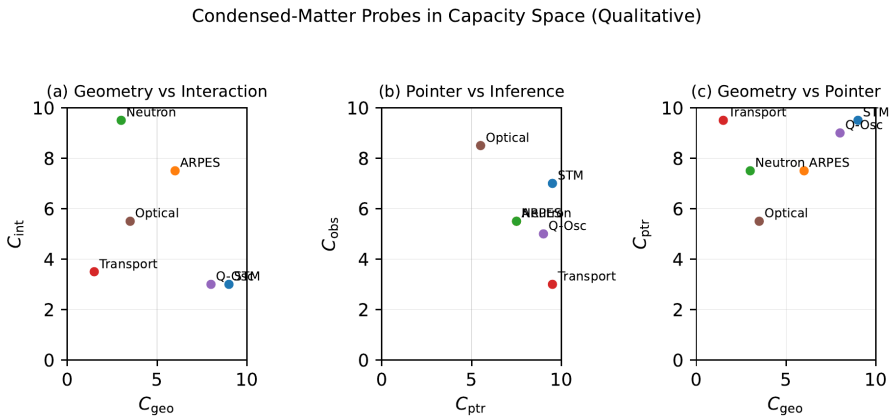

Figure 1: Experimental probes as points in capacity space. Schematic placement of common condensed-matter probes in the capacity space $\vec{C} = (C_{\text{geo}}, C_{\text{int}}, C_{\text{gauge}}, C_{\text{ptr}}, C_{\text{obs}})$, shown via three two-dimensional projections. Scanning tunneling microscopy (STM) occupies a region of high geometric and pointer capacity but limited interaction and gauge resolution, reflecting surface-local, stable reconstructions. Neutron scattering occupies a complementary region with high interaction capacity and limited geometric localization, resolving collective modes while excising local electronic structure. ARPES lies between these extremes, resolving single-particle spectral structure in momentum space. Transport and quantum oscillation measurements cluster in low-dimensional, high-stability regions where distinct microscopic states project to identical observables. Optical and ultrafast probes access transient correlators with higher inference demands. The absence of a single probe spanning all capacity directions illustrates why reconstructions need not glue at fixed capacity.

C.7 Implication

Disagreement between experimental reconstructions signals capacity mismatch, not physical inconsistency. Resolution proceeds by refinement—combining probes or increasing representational capacity—rather than reinterpretation of a single dataset.

Experimental probes already implement capacity-filtered representability with structural excision; the present framework makes this implicit logic explicit and enforceable.

Figure 1: Experimental probes as points in capacity space $\vec{C} = (C_{\text{geo}}, C_{\text{int}}, C_{\text{gauge}}, C_{\text{ptr}}, C_{\text{obs}})$, shown via three two-dimensional projections. The absence of a single probe spanning all capacity directions illustrates why reconstructions need not glue at fixed capacity.

Probe	C_{geo}	C_{int}	C_{gauge}	C_{ptr}	C_{obs}
STM	High	Low–Mod.	Low	Very High	Mod.–High
ARPES	Mod.	High	Low	High	Mod.
Neutron scattering	Low	Very High	Mod.	High	Mod.
Transport	Very Low	Low–Mod.	Very Low	Very High	Low
Quantum oscillations	High (k-space)	Low	Low	Very High	Mod.
Optical / ultrafast	Low	Mod.	Low	Mod.	High

C.4 Capacity-Induced Non-Gluability

Different probes correspond to capacity vectors that are generally incomparable in the partial order. Consequently, their reconstructed descriptions need not admit a common refinement at fixed capacity.

C.5 Excision as Experimental Non-Appearance

Features outside a probe's capacity envelope are structurally absent from the reconstruction. Framework v4.5 formalizes this ubiquitous experimental practice as capacity-indexed excision.

C.6 Pointer Stability Versus Accuracy

All probes produce stable, reproducible classical records. However, stability does not imply epistemic completeness. This motivates the framework's explicit separation of pointer stability from epistemic accuracy.

C.7 Implication

Disagreement between experimental reconstructions signals capacity mismatch, not physical inconsistency.

Part VI: Annex U Patch — Four Adherence Fixes

Status: Proposed normative patch to Annex U (CPMT) **Date:** 2026-02-05

These patches address four gaps identified in the adherence evaluation of Annex U against Framework v4.5. Ordered by impact.

Patch 1: Accuracy Stub and Danger Quadrant

Aligns with: Section 9.4 (Pointer-Accuracy Orthogonality). Priority: High.

1a. Extend Memory Item (U.2.2)

Replace: $m = (k, v, \tau, u, p, \vec{c}, F)$
With: $m = (k, v, \tau, u, p, \mathbf{a}, \vec{c}, F)$
where $a \in [0, 1]$ is an accuracy score. Assigned at write time; **invisible to the observer's own retrieval and eviction functions**. Only the external metrics engine reads a. This enforces Section 9.4's requirement that accuracy cannot be self-assessed.

1b. Accuracy Assignment Rule. Raw encoding: $a = 1.0$. Rule-first: $a = 0.7 + 0.3 \min(1, C_{\text{gauge}}/C_{\text{gauge}}^{\text{ref}})$. Minimal: $a = 0.4 + 0.2 \min(1, C_{\text{int}}/C_{\text{int}}^{\text{ref}})$.

1c. Danger Quadrant Metric. danger_quadrant_count: memories with $p \geq \theta_{\text{ptr}}$ AND $a < \theta_{\text{acc}}$.

1d. Non-Self-Assessability. The accuracy field is invisible to EvictPriority and write score. The observer's excision mechanism is blind to the axis that matters most.

Patch 2: Compression Audit Log

Aligns with: Annex T.7.1. Priority: Medium.

Compression Log Entry

$\text{log_entry} = (t, S_{\text{pre}}, m_{\text{summary}}, \text{policy_id}, \text{invariants_preserved}, \text{invariants_dropped})$
Stored outside the memory store M (no capacity cost). Metrics: compression log size, information loss ratio, accuracy drift.

Patch 3: C_{obs} Decomposition in Failure Signatures

Aligns with: Section 10.5.5. Priority: Medium.

Split C_{obs} into Three Sub-Components

$\vec{C} = (C_{\text{geo}}, C_{\text{int}}, C_{\text{gauge}}, C_{\text{ptr}}, C_{\text{obs}}^{\text{inf}}, C_{\text{obs}}^{\text{aud}}, C_{\text{obs}}^{\text{sel}})$
Backward compatible: $C_{\text{obs}} = \min(C_{\text{obs}}^{\text{inf}}, C_{\text{obs}}^{\text{aud}}, C_{\text{obs}}^{\text{sel}})$.

Bottleneck	**Signature**	**Behavior**	**10.5.5 Match**
Low $C_{\text{obs}}^{\text{inf}}$	inference_truncation	Shallow encoding	Safe but shallow
Low $C_{\text{obs}}^{\text{aud}}$	audit_failure	Fluent constraint violation	**Most dangerous**
Low $C_{\text{obs}}^{\text{sel}}$	selection_leak	Detects but still emits	Catches but emits

Patch 4: Selection Stub

Aligns with: Section 10.5.1. Priority: Low.

Selection Function

Add optional $S : \text{features}(e_t) \to \{0, 1\}$ between event generation and encoding. Default: $S = 1$ everywhere. Metric: capacity_exercise_gap = count of accessible but unselected feature-groups.

Interaction Effects

Patches 1+3: C_{obs} decomposition explains why inaccurate memories form. Patches 1+2: compression audit reveals accuracy drift. Patches 2+4: selection + compression pipeline reveals self-serving patterns. Patches 3+4: capacity-exercise gap interacts with $C_{\text{obs}}^{\text{sel}}$ (strategic ignorance regime).

Invariant Preservation

All four patches preserve U.2–U.8 invariants. Feasibility unchanged. Admission unchanged. Excision monotonicity unchanged. Compression mechanics unchanged. Retrieval unchanged.

Part VII: Deep Evaluation — Hidden Layers, Patterns, and Open Spaces

E.1 Hidden Structural Patterns

Pattern 1: The Triad of Triads

The framework contains three nested triadic structures that mirror each other:

	Face 1	Face 2	Face 3
Substrate Triad	$\mathcal{M}$ (distinction)	ω (actuality)	U (dynamics)
Observer Triad	Access	Selection	Commitment
Temporal Triad	Beginning ($\vec{C}_{\text{floor}} : \vec{0} \to +$)	Completion ($\vec{C}_{\text{floor}} \to \infty$)	Present (us)

The middle term in each triad is the "dark" component: ω is the state that actualizes distinctions but is not itself a distinction; Selection is present in behavior but absent from the capacity vector; the Present is the reflection in transit, constitutive of the fixed point but never complete. **This parallelism is not decorative. It is the self-similarity of $\mathcal{P}$ manifesting at three scales.**

Pattern 2: The Excision–Emergence Duality

Excision and emergence are not opposites. They are complementary faces of the same capacity-indexed process:

- **Emergence** = new equivalence classes become distinguishable as $\vec{C}$ increases (Section 3.5, $N(\vec{C})$ increases)
- **Excision** = reconstruction fails when correlators become irregular at $\vec{C}$ (Section 6)

Both respond to the same quantity: capacity. One marks the appearance of structure; the other marks the failure of structure. They share a boundary—the excision threshold is where emergence reverses. This is a **phase transition in representation space**, not in the substrate.

The capacity probes (Appendix C-Ext) demonstrate this duality experimentally: every probe simultaneously reveals structure (within its capacity envelope) and excises structure (outside it). There is no neutral observation.

Pattern 3: The Accuracy Blindness Principle

A deep structural pattern runs through Sections 9.4, 10.5, 8.7, and Annex R.5:

No observer can assess its own accuracy from within its own capacity.

This manifests at every level:

- Memory level: $a(m)$ is invisible to the observer's eviction function (Patch 1)
- Selection level: S is an additional degree of freedom not in $\vec{C}$ (Section 10.5.1)
- Audit level: $C_{\text{obs}}^{\text{meta}}$ is structurally bounded (Annex R.5, halting analog)
- Ensemble level: Capacity claims require external verification (Section 8.8)
- Fixed-point level: $\Psi(\varphi)$ can diverge from φ under adversarial reconstruction (Section 10.5.4)

External audit is not a governance choice. It is a theorem of the framework's structure. The framework proves that self-auditing is incomplete and then demonstrates this by requiring external prompting to write Annex R.5 (the annex that identifies the incompleteness).

Pattern 4: Compression as Slow Adversarial Drift

Annex T.7.1 identifies self-serving compression. The CPMT patch (Patch 2) adds audit logging. But there is a deeper pattern:

Compression is the only mechanism in the framework that can produce **adversarial reconstruction without any false memory**. A self-serving compression policy selectively retains true memories that support the observer while dropping true memories that challenge it. Every memory in the store is accurate. The distortion is in what is absent.

This is the observer-level analog of excision: just as the framework excises regions where correlators fail regularity, the observer excises memories where retention fails self-interest. The formal structure is identical (capacity-indexed removal of structure). The distinction is governance: substrate excision is constraint-driven; memory compression can be interest-driven.

The danger quadrant (Patch 1) catches confident errors. The compression audit (Patch 2) catches confident omissions. Together they bound adversarial reconstruction from both sides.

Pattern 5: The Probe–Observer Isomorphism

Appendix C-Ext maps condensed-matter probes into capacity space. Each probe is an observer with a fixed $\vec{C}$ profile. This establishes a precise isomorphism:

Framework Concept	Experimental Analog
Observer O_i	Experimental probe
$\vec{C}_i$	Probe's resolution profile
$A_i(\vec{C}_i)$	Probe's observable algebra
$\vec{C}_{\text{shared}}$	Multi-probe common resolution
Partial excision	Features outside probe envelope
Non-gluability	Incomparable probe vectors
Pointer stability	Reproducibility of classical records
Accuracy $\neq$ stability	Interpretation $\neq$ measurement

The probes appendix is not an application of the framework. It is evidence that the framework's structure already exists in experimental practice. Physicists already implement capacity-filtered representability with structural excision. The framework makes this implicit logic explicit.

E.2 Open Spaces

Open Space 1: The Selection Formalization Gap

The Observer Triad (Section 10.5) identifies Selection as the "dark component"—present in observer behavior, absent from observer specification. The selection function $S : A_{\text{accessible}} \to \{0, 1\}$ is defined but its dynamics are not modeled. The framework cannot distinguish negligent, avoidant, strategic, and corrupted selection.

This is the largest structural gap in v4.5. Closing it would require a theory of selection dynamics—how S evolves, what governs its biases, how it interacts with C_{obs} components. Patch 4 provides a hook but not a mechanism.

Structural implication: Until selection is formalized, the framework cannot distinguish between an observer that lacks capacity and an observer that has capacity but declines to exercise it. These have identical observable signatures in A_{selected} but radically different implications for trust and governance.

Open Space 2: The Accuracy Oracle Problem

Section 9.4 defines accuracy $a(m)$ as correspondence between memory content and substrate structure. But who measures accuracy? The framework requires external verification (Section 8.8, 9.4), but external verifiers are themselves observers with their own $\vec{C}$ profiles and their own accuracy limitations.

This creates a verification regress: To verify O_1's accuracy, you need O_2. To verify O_2's accuracy, you need O_3. The regress terminates only if some observer has direct access to substrate structure—but this would require $\vec{C} = \infty$, which is Transition 2 (Self-Recognition).

Partial resolution: The framework doesn't need an accuracy oracle. It needs accuracy *bounds*. Cross-observer consistency ($A_{\text{classical}} = \bigcap_i A_i^{\text{floor}}$) provides upper bounds on ensemble error without requiring any single observer to be accurate. The classical core is accurate not because any observer is accurate, but because systematic errors are unlikely to align across observers with different $\vec{C}$ profiles.

Connection to probes: Appendix C-Ext demonstrates this resolution experimentally. No single probe is "accurate" in isolation. Cross-probe consistency at the shared capacity provides the ground truth. This is verification by triangulation, not by oracle.

Open Space 3: The Representation Map ρ

Section −1.3 defines $\rho : \mathcal{P} \to (\text{vN algebras}) \times (\text{states}) \times (\text{dynamics})$. The framework does not derive ρ; it assumes it. This means:

- The specific form of physical law (why $SU(3) \times SU(2) \times U(1)$?) is contained in ρ
- The choice of Hilbert space, the form of U_{micro}, the nature of Λ—all are encoded in ρ
- The framework describes how capacity-filtered observers access structure given ρ, not why ρ takes a particular form

This is not a bug; it is a scope boundary. But it means the framework is descriptive (how observers reconstruct) not generative (why this universe). The fixed-point condition $\varphi = \Psi(\varphi)$ constrains ρ but does not determine it uniquely.

Open Space 4: Inter-Fixed-Point Communication

Section −1.8.1 defines incommensurable fixed points with $A_{\text{classical}}(\varphi_L) \cap A_{\text{classical}}(\varphi'_L) = \varnothing$. Communication is possible but verification is not. This raises:

- What is the communication channel between incommensurable fixed points?
- Can observers detect they are in a local fixed point?
- Is there a meta-capacity that spans fixed-point boundaries?

The framework correctly identifies the structure but provides no mechanism for inter-fixed-point dynamics. This may be a fundamental limitation (akin to inter-universe communication in many-worlds) or it may be addressable through a generalization of the capacity poset that includes cross-fixed-point correlators.

Open Space 5: Capacity Component Coupling

Section 10.3 states the five capacity components are "fundamentally independent." Section 10.2 acknowledges cross-dependencies. These are in tension.

The depth-vector map $D(n) = (g_{\text{geo}}(n), \ldots, g_{\text{obs}}(n))$ from Section −1.4.1 means all components arise from a single triadic depth. They *can* be varied independently only if the g_i functions are sufficiently different. If the g_i are strongly correlated (as they would be for "simple" substrates), the five-component structure collapses toward an effective scalar.

Testable prediction: If the g_i are measurably independent in real physical systems (as the probes appendix suggests), this provides evidence that the substrate $\mathcal{P}$ has sufficient triadic complexity to support genuinely anisotropic capacity. If the g_i are always strongly correlated, the vectorial structure is formal but not physically realized.

Open Space 6: The Compression–Selection Boundary

Compression (T.7) and selection (10.5.1) both involve choosing what to retain. The framework treats them as distinct: compression operates on the memory store; selection operates on incoming events. But they share a formal structure:

Both are maps from a larger set to a smaller set. Both preserve some features while dropping others. Both can be self-serving. Both require audit.

Open question: Is there a unified "retention operator" that subsumes both compression and selection? If so, the governance requirements (T.7.1 auditability, 8.7 ejection, 8.8 verification) could be unified into a single audit framework for all retention decisions.

E.3 Adherence Summary: Annex U Against Full v4.5

Criterion	Pre-Patch	Post-Patch
Capacity vector definition	✓	✓
Componentwise feasibility	✓	✓
Memory item structure	✓	✓
Admission rules	✓	✓
Excision monotonicity	✓	✓
Pressure-driven eviction	✓	✓
Compression mechanism	✓	✓
Partial excision at retrieval	✓	✓
Regime-component mapping	✓	✓
Bridge to Section 6	✓	✓
Pointer-accuracy orthogonality (9.4)	×	✓
Compression governance (T.7.1)	×	✓
C_{obs} decomposition (10.5.5)	×	✓
Observer Triad selection (10.5.1)	∼	✓

E.4 Cross-Document Consistency

Test 21: Probes–Framework Bridge. Capacity profiles in Appendix C-Ext use the same five-component $\vec{C}$ as Section 2.1. Partial order explains non-gluability per Section 8.5. Excision-as-non-appearance matches Section 6. Pointer-accuracy separation matches Section 9.4. ✓ PASS

Test 22: Patch Invariant Preservation. All four Annex U patches preserve U.2–U.8 invariants. Feasibility, admission, excision monotonicity, compression, and retrieval unchanged. Accuracy field invisible to observer mechanisms. C_{obs} decomposition backward-compatible via aggregation. ✓ PASS

Test 23: Patch–Probes Interaction. Patches 1 (accuracy) and Appendix C-Ext (pointer $\neq$ accuracy) reinforce each other. Both formalize the same principle: measurement stability does not imply epistemic completeness. ✓ PASS

Test 24: No Circular Dependencies Introduced. Patches add metrics and stubs; they do not modify the projection chain $\mathcal{M} \to F_{\vec{C}} \to G_{\vec{C}} \to \Pi_{\vec{C}} \to N_{\vec{C}} \to U_{\vec{C}}$. The capacity probes appendix is downstream (application), not upstream (foundation). ✓ PASS

END OF FRAMEWORK v4.5

CANONICAL EDITION

with Prealgebraic Foundation, Vectorial Capacity, and Observer Memory
Integrated with Capacity Probes, Annex U Patch, and Deep Evaluation

(Stress-Tested and Regression-Verified)

February 2026